26900414 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEPHY305 - INTRODUCTION TO GROUP THEORY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Give an example of a direct product group. [U] / [CO1]
2. Does the set of positive integers form a group with addition as the rule of composition? Answer with reason. [A] / [CO1]
3. Identify a proper subgroup of cyclic group C_4. [A] / [CO1]
4. Give the symbol and order of the complete symmetry group of a regular tetrahedron. [K] / [CO2]
5. The order of D_{nd} is $4n$. Find the order of D_{3d} and the number of its proper rotations. [A] / [CO2]
6. A cube has 48 operations in its complete symmetry group O_h, of which 24 are proper rotations. How many are improper operations? [A] / [CO2]
7. When are two mirror planes called equivalent? State the relation between their reflection operations. [K] / [CO2]

8. A representation of C_2 has characters $\chi(E) = 2$ and $\chi(s) = 0$. Use the character criterion to decide whether it is reducible. [A] / [CO1, CO3]

9. In an orthonormal basis, let $\hat{P} = |e_1\rangle\langle e_1|$. Find $\hat{P}(2|e_1\rangle + 3|e_2\rangle)$. [A] / [CO3]

10. State the row orthogonality relation for irreducible characters of a finite group of order h, including complex conjugation. [U] / [CO1, CO3]

11. Determine whether the following matrix is hermitian : $\begin{pmatrix} 3 & 1+i \\ 1-i & 7 \end{pmatrix}$. [A] / [CO4]

12. Write down two examples of continuous groups. [U] / [CO4]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

13. Explain factor groups with the help of a suitable example. [U] / [CO1]

14. Find the conjugacy classes of permutation group S_4. [A] / [CO1]

15. Name the proper rotation groups associated with the five convex regular polyhedra. For each distinct group, give its symbol, its order, and the polyhedron or polyhedra that possess it. [U] / [CO2]

16. In D_2, the half-turns act as $C_{2x}(x, y, z) = (x, -y, -z)$ and $C_{2y}(x, y, z) = (-x, y, -z)$. For $P = (1, 2, 3)$: (a) apply C_{2x}; (b) then apply C_{2y}; (c) identify the single rotation equivalent to this pair of operations. [A] / [CO1, CO2]

17. In $\mathbb{C}[C_2]$, the group is $C_2 = \{E, s\}$ with $s^2 = E$. (a) Expand $(E + 2s)(3E - s)$. (b) In the group-algebra basis $(|E\rangle, |s\rangle)$, write the matrices for left multiplication by E and s. (c) Find the characters of this regular representation and its dimension. [A] / [CO1, CO3]

18. State (a) the relation between the number of inequivalent irreducible complex representations and conjugacy classes, (b) the sum-of-squares theorem for their dimensions, and (c) the consequence for an abelian group of order h. [U] / [CO1, CO3]

19. Define a continuous group. Provide two examples. [U] / [CO4]

20. Write down thas Casimir operator for $SU(2)$ group. Show that representations of $SU(2)$ are labelled by eigenvalues of Casimir and L_3 . [U] / [CO4, CO5]

[6x5 = 30]

Part C

Problem type Questions

Answer any **4** questions

Each question carries **5** marks

21. Obtain the invariant subgroups of permutation group S_3 . [A] / [CO1]

22. Let $\mathbb{Z}$ be the group of integers under addition. If $\mathbb{Z}_+ \subset \mathbb{Z}$ is its subgroup of even integers, find the left cosets and the index of $\mathbb{Z}_+$ in $\mathbb{Z}$. [A] / [CO1]

23. For $P = (1, 2, 3)$, find its image under each operation separately: (a) C_2 about the z-axis; (b) reflection in the xz-plane; (c) inversion through the origin. State the coordinate rule in each case. [A] / [CO2]

24. Let $D(s)|e_1\rangle = |e_2\rangle$ and $D(s)|e_2\rangle = |e_1\rangle$, with $D(E) = I$. Take $|+\rangle = (|e_1\rangle + |e_2\rangle)/\sqrt{2}$ and $|-\rangle = (|e_1\rangle - |e_2\rangle)/\sqrt{2}$. (a) Find their images under $D(s)$. (b) Write $D(s)$ in the ordered basis $(|+\rangle, |-\rangle)$. (c) Give the two one-dimensional character rows and explain reducibility. [A] / [CO1, CO3]

25. Let $|e_1\rangle, |e_2\rangle$ be orthonormal and $|f_+\rangle = (|e_1\rangle + |e_2\rangle)/\sqrt{2}$, $|f_-\rangle = (|e_1\rangle - |e_2\rangle)/\sqrt{2}$. For $|v\rangle = 3|e_1\rangle + |e_2\rangle$, (a) calculate $\langle f_+|v\rangle$ and $\langle f_-|v\rangle$; (b) write its expansion in the f basis; (c) check its squared norm in both bases. [A] / [CO3]

26. Show that a rotation about X axis does not commute with a rotation about Y axis. [U] / [CO4, CO5]

27. Verify whether the matrix $\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}$ is an element of $SO(3)$ group. [A] / [CO5]

[5x4 = 20]